WORKED SOLUTION

STEP III 2009, Question 1

Pure • Coordinate Geometry of Circles

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Read the question before you go on. It is in the STEP Support Programme's question database:

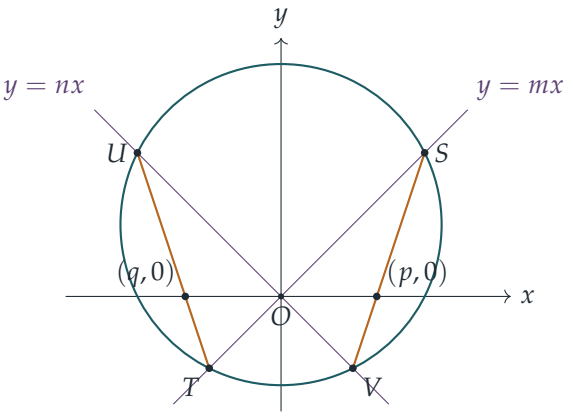
step.maths.org/questions/09-s3-q1

The key idea

All four points lie on lines through the origin, so each is described by a single number: its x-coordinate. Intersecting a circle with the line $y = mx$ gives a quadratic whose two roots are s and t, and we never need the roots themselves, only their sum and product. The final identity then comes from grouping the numerator of $p + q$ so that only st, $s + t$, uv and $u + v$ appear.

Solution

Throughout, S and T lie on the line $y = mx$ and U and V lie on $y = nx$, both lines through the origin O. We assume $m \neq n$ (otherwise all four points are on one line), and that SV and UT are not horizontal, so that they do meet $y = 0$; that is, $ms \neq nv$ and $mt \neq nu$.



The intercepts p and q

The line through $S(s, ms)$ and $V(v, nv)$ has gradient $\dfrac{nv - ms}{v - s}$, so its points are

$$y - ms = \frac{nv - ms}{v - s}(x - s).$$

Putting $y = 0$ and $x = p$ gives $p - s = -\dfrac{ms(v - s)}{nv - ms}$, so

$$p = \frac{s(nv - ms) - ms(v - s)}{nv - ms} = \frac{nsv - msv}{nv - ms} = \frac{(m - n)sv}{ms - nv},$$

as required. (If $v = s$ the line SV is vertical, so $p = s$; the formula still gives $\frac{(m-n)s^2}{(m-n)s} = s$.)

The line UT joins $T(t, mt)$ on $y = mx$ to $U(u, nu)$ on $y = nx$, exactly as SV joins S to V. Replacing s by t and v by u,

$$q = \frac{(m - n)tu}{mt - nu}.$$

The quadratic for s and t

A point (x, mx) lies on $x^2 + (y - c)^2 = r^2$ exactly when $x^2 + (mx - c)^2 = r^2$, that is,

$$(1 + m^2)x^2 - 2mc\,x + c^2 - r^2 = 0.$$

Both s and t satisfy this. Taking S and T to be the two (distinct) intersections of the line with the circle, s and t are the two roots, so by the relations between roots and coefficients,

$$st = \frac{c^2 - r^2}{1 + m^2}, \qquad s + t = \frac{2mc}{1 + m^2}.$$

Showing $p + q = 0$

The same argument on the line $y = nx$ shows that u and v are the roots of $(1 + n^2)x^2 - 2nc\,x + c^2 - r^2 = 0$, so

$$uv = \frac{c^2 - r^2}{1 + n^2}, \qquad u + v = \frac{2nc}{1 + n^2}.$$

Now add the two intercepts over a common denominator:

$$p + q = (m - n)\,\frac{sv(mt - nu) + tu(ms - nv)}{(ms - nv)(mt - nu)}.$$

Ask yourself How do we tame the numerator, which involves all four of s, t, u, v?

Expand it and pair terms by their coefficient m or n:

$$\begin{aligned} sv(mt-nu)+tu(ms-nv) &= m\,stv - n\,suv + m\,stu - n\,tuv \\ &= m\,st(u+v) - n\,uv(s+t). \end{aligned}$$

Every term is now a symmetric function we know. Substituting,

$$\begin{aligned} m\,st(u+v) - n\,uv(s+t) &= m\cdot\frac{c^2-r^2}{1+m^2}\cdot\frac{2nc}{1+n^2} - n\cdot\frac{c^2-r^2}{1+n^2}\cdot\frac{2mc}{1+m^2} \\ &= \frac{2mnc(c^2-r^2)}{(1+m^2)(1+n^2)} - \frac{2mnc(c^2-r^2)}{(1+m^2)(1+n^2)} = 0. \end{aligned}$$

The denominator $(ms-nv)(mt-nu)$ is non-zero by our assumption, so $p+q=0$, as required.

Checks

Take $c=1$, $r^2=5$, $m=1$, $n=-1$, as in the figure. The quadratics are $x^2-x-2=0$ and $x^2+x-2=0$, so $s=2$, $t=-1$ and $u=-2$, $v=1$. Then $st=-2=\frac{1-5}{2}$ and $s+t=1=\frac{2}{2}$, as predicted. The line from $S(2,2)$ to $V(1,-1)$ has gradient 3 and crosses $y=0$ at $p=\frac{4}{3}$, matching $\frac{2\cdot2\cdot1}{2+1}$; the line from $U(-2,2)$ to $T(-1,-1)$ crosses at $q=-\frac{4}{3}$. A computer check with 16 000 random circles and lines gave $|p+q|$ below 10^{-12} every time.

Common slips

- Pairing the wrong points: q comes from U and T, so the swap is $s\to t$, $v\to u$; swapping $s\leftrightarrow v$ instead gives p again.
- Trying to solve the quadratics for s,t,u,v explicitly. The square roots make the algebra hopeless; the grouping $m\,st(u+v)-n\,uv(s+t)$ is what makes it collapse.

Beyond STEP

The line $y=0$ meets the circle in a chord whose midpoint is O, since the centre $(0,c)$ is on the y-axis. So this question proves the *butterfly theorem*: if two chords ST and UV pass through the midpoint O of a chord, then SV and UT cut that chord at points equidistant from O. Our proof never used $c^2<r^2$, so it even works when O is outside the circle, which is a hint that the natural setting is projective geometry, where the result is a case of Desargues' involution theorem. Coxeter and Greitzer's *Geometry Revisited* is a lovely place to read more.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.